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This Small Device Could Stop Your Laundry From Polluting the Ocean With Microplastics

Bristol-based startup Matter has built an add-on filter for washing machines that captures 97 percent of microplastics shed from synthetic clothing. The European Union is now preparing rules that could require every new washing machine to come fitted with a similar filter.

By VokaMedia Reporter (Intern)

BRISTOL — About 60 percent of the fabric produced today is synthetic, meaning it is essentially made of plastic. Every time clothing made from this material is washed, the machine gradually wears down the fibres and releases tiny particles that ordinary filters cannot effectively catch.

Adam Root, CEO of Matter, describes the process as similar to a cheese grater working inside the washing machine, slowly scraping away at the plastic until it breaks down into microscopic particles that eventually flow into the water system.

The scale of the problem is considerable. According to Matter's research, roughly a gram of microplastic is released with every wash. In the UK, where around 24 million households own a washing machine, that adds up to an estimated 16 tonnes entering the water supply every day. Across Europe, with more than 100 million households, the figure is naturally far higher.

Root explains that once microplastics reach the water, they do not simply dissolve. Instead, the particles act much like a pill, carrying harmful chemicals into the organisms that absorb them, including phytoplankton and zooplankton, which form the base of the food chain and produce most of the world's oxygen.

Standard washing machine filters typically have holes around five millimetres wide, designed to stop coins or buttons from damaging the pump rather than to catch microplastics. Matter's device works differently: it is fitted as an add-on to an existing washing machine, with no need to replace the appliance itself, and installation is said to take under 10 minutes. In testing, the filter captured 97 percent of particles as small as 10 microns, roughly one-sixth the width of a human hair.

In everyday use, a small indicator light switches on when the filter needs emptying, usually about once a month. Matter is currently selling its product in 11 countries, with plans to expand to 22 by the end of this year and eventually across the entire EU market, backed by partnerships with several major washing machine brands.

For now, the collected microplastic can legally be sent to landfill under existing EU rules. Matter, however, is pushing for a greener alternative through a programme called Love Your Lint, which collects material from users to trial turning it back into new materials, although this has not yet reached commercial scale. The company was also a finalist for the 2025 Earthshot Prize, the environmental award championed by Prince William, which has helped raise its profile on the global stage.

The EU itself is now preparing new legislation that would require every new washing machine sold to include a microplastic filter, a move that could significantly expand the market for solutions like Matter's.

Tags: microplastics, water pollution, green innovation, washing machines, British technology

Caption: Adam Root, CEO of Matter, showing the microplastic filter he built for washing machines. Source: Denis Loctier / Euronews